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Agricultural systems having stalk sensors and data visualization systems and related devices and methods

Abstract

The disclosure relates to agricultural systems. The systems have stalk sensor assemblies and/or data visualization systems. The stalk sensor assemblies are configured for assessing the size and other characteristics about crops, such as corn and other grains entering an agricultural implement, such as a harvester. The stalk sensor assemblies may use an estimation of the stalk perimeter to establish stalk size and therefore further features about the crop. The visualization system utilizes data from the stalk sensor assemblies to calculate and display relevant information about the crop.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2419061	12/1946	Emery	N/A	N/A
2813709	12/1956	Hyman	N/A	N/A
3038544	12/1961	Richey et al.	N/A	N/A
3178873	12/1964	Meyer	N/A	N/A
3246701	12/1965	Horst	N/A	N/A
3271940	12/1965	Robert	N/A	N/A
3616690	12/1970	Harden	N/A	N/A
3780817	12/1972	Videon	N/A	N/A
3853000	12/1973	Barnett	N/A	N/A
3972381	12/1975	Gail	N/A	N/A
3992933	12/1975	Randolph	N/A	N/A
4121049	12/1977	Roeber	N/A	N/A
4126984	12/1977	Gail	N/A	N/A
4166349	12/1978	Coenenberg et al.	N/A	N/A
4197690	12/1979	Eistert	N/A	N/A
4295323	12/1980	Maier et al.	N/A	N/A
4362218	12/1981	Shoberg	N/A	N/A
4418521	12/1982	Orlando	N/A	N/A
4423640	12/1983	Jetter	N/A	N/A
4505094	12/1984	Demorest	N/A	N/A
4528804	12/1984	Williams	N/A	N/A
4751849	12/1987	Paros	N/A	N/A
4883964	12/1988	Bohman	N/A	N/A
4896491	12/1989	Warnsholz	N/A	N/A
4918441	12/1989	Bohman	N/A	N/A
4967362	12/1989	Schutten et al.	N/A	N/A
5044210	12/1990	Kuhn et al.	N/A	N/A
5216795	12/1992	Hansson	N/A	N/A

5264709	12/1992	Kamimura	N/A	N/A
5343761	12/1993	Myers	N/A	N/A
5369603	12/1993	Myers	N/A	N/A
5463854	12/1994	Chmielewski, Jr.	N/A	N/A
5480354	12/1995	Sadjadi	N/A	N/A
5568405	12/1995	Easton et al.	N/A	N/A
5598794	12/1996	Harms et al.	N/A	N/A
5680750	12/1996	Stefl	N/A	N/A
5751576	12/1997	Monson	239/161	A01B 79/005
5790428	12/1997	Easton et al.	N/A	N/A
5847290	12/1997	Kim	N/A	N/A
5878561	12/1998	Gunn	N/A	N/A
5921070	12/1998	Chamberlain	N/A	N/A
5927054	12/1998	Chamberlain	N/A	N/A
5991694	12/1998	Gudat et al.	N/A	N/A
6041583	12/1999	Goering	N/A	N/A
6073427	12/1999	Nichols	460/7	A01D 41/1271
6085846	12/1999	Buchl et al.	N/A	N/A
6119442	12/1999	Hale	N/A	N/A
6202395	12/2000	Gramm	N/A	N/A
6216795	12/2000	Buchl	N/A	N/A
6272819	12/2000	Wendte	N/A	N/A
6431981	12/2001	Shinners	N/A	N/A
6530197	12/2002	Christensen	N/A	N/A
6584390	12/2002	Beck	N/A	N/A
6615570	12/2002	Beck	N/A	N/A
6668223	12/2002	Blackmore et al.	N/A	N/A
6774320	12/2003	Simons	N/A	N/A
6983582	12/2005	Muckler	N/A	N/A
6986582	12/2005	Muckler	N/A	N/A
7354341	12/2007	Smith	N/A	N/A
7357036	12/2007	Steprath	N/A	N/A
7401506	12/2007	Kunow	N/A	N/A
7401528	12/2007	Deppermann et al.	N/A	N/A
7647753	12/2009	Schlipf	N/A	N/A
7716905	12/2009	Wilcox et al.	N/A	N/A
7739861	12/2009	Mackin	N/A	N/A
7790991	12/2009	Verhaeghe	N/A	N/A
7916898	12/2010	Anderson	N/A	N/A
8006472	12/2010	Schreiner	N/A	N/A
8010261	12/2010	Brubaker	N/A	N/A
8196380	12/2011	Carboni	N/A	N/A
8201388	12/2011	Vandeven	N/A	N/A
8215191	12/2011	Tragesser et al.	N/A	N/A
8220235	12/2011	Kowalchuk	N/A	N/A
8224534	12/2011	Kowalchuk	N/A	N/A
8418636	12/2012	Liu et al.	N/A	N/A
8820039	12/2013	Werning	N/A	N/A

8987615	12/2014	Khataavkar	N/A	N/A
9030549	12/2014	Redden	N/A	N/A
9064173	12/2014	Redden	N/A	N/A
9066463	12/2014	Lange	N/A	N/A
9213905	12/2014	Lange	N/A	N/A
9232693	12/2015	Hendrickson et al.	N/A	N/A
9282693	12/2015	Anderson	N/A	N/A
9310329	12/2015	Acheson et al.	N/A	N/A
9320196	12/2015	Dybro et al.	N/A	N/A
9322629	12/2015	Sauder et al.	N/A	N/A
9360383	12/2015	Coleman	N/A	N/A
9372109	12/2015	Acheson et al.	N/A	N/A
9410840	12/2015	Acheson et al.	N/A	N/A
9423249	12/2015	Deppermann et al.	N/A	N/A
9485914	12/2015	Schleicher et al.	N/A	N/A
9565802	12/2016	Schleicher	N/A	N/A
9578804	12/2016	Gessel	N/A	N/A
9578808	12/2016	Dybro	N/A	N/A
9609806	12/2016	Schlipf	N/A	N/A
9645006	12/2016	Phelan	N/A	N/A
9658201	12/2016	Redden et al.	N/A	N/A
9668412	12/2016	Ritter	N/A	N/A
9668420	12/2016	Anderson	N/A	N/A
9693496	12/2016	Tevs et al.	N/A	N/A
9693503	12/2016	Dybro et al.	N/A	N/A
9696162	12/2016	Anderson	N/A	N/A
9717171	12/2016	Redden et al.	N/A	N/A
9756771	12/2016	Redden	N/A	N/A
9804097	12/2016	Tang et al.	N/A	N/A
9826673	12/2016	Ray	N/A	N/A
9832928	12/2016	Dybro et al.	N/A	N/A
9867334	12/2017	Jongmans et al.	N/A	N/A
9867335	12/2017	Obbink et al.	N/A	N/A
9894835	12/2017	Sauder et al.	N/A	N/A
9921064	12/2017	Schleicher	N/A	N/A
9927242	12/2017	Schleicher	N/A	N/A
9936631	12/2017	Hubner et al.	N/A	N/A
9936637	12/2017	Anderson et al.	N/A	N/A
9972058	12/2017	Romier	N/A	N/A
9974233	12/2017	Ueda et al.	N/A	N/A
10034424	12/2017	Anderson	N/A	N/A
10039228	12/2017	Walker	N/A	A01D 63/00
10039231	12/2017	Anderson	N/A	G01B 21/08
10130035	12/2017	Crow	N/A	A01D 45/023
10178828	12/2018	Hendrickson et al.	N/A	N/A
10188037	12/2018	Sauder	N/A	N/A
10255670	12/2018	Wu	N/A	N/A

10295703	12/2018	Dybro	N/A	N/A
10299422	12/2018	Schleicher	N/A	N/A
10473592	12/2018	Kramer et al.	N/A	N/A
10537060	12/2019	Sauder et al.	N/A	N/A
10582662	12/2019	Ricketts	N/A	A01D 45/021
10713768	12/2019	Berghoefer	N/A	N/A
10820508	12/2019	Dix et al.	N/A	N/A
10859479	12/2019	Brune et al.	N/A	N/A
11048938	12/2020	Hundley et al.	N/A	N/A
11064653	12/2020	Zielke	N/A	N/A
11079725	12/2020	Palla	N/A	N/A
11129333	12/2020	Derscheid	N/A	N/A
11234366	12/2021	Darr	N/A	N/A
11275941	12/2021	Papanikolopoulos et al.	N/A	N/A
11297768	12/2021	Schildroth	N/A	N/A
11432464	12/2021	Hayashi	N/A	N/A
11678607	12/2022	Zielke	N/A	N/A
11704810	12/2022	Sneyders	N/A	N/A
11758845	12/2022	Zielke	N/A	N/A
11758848	12/2022	Eichhorn	N/A	N/A
2002/0173893	12/2001	Blackmore et al.	N/A	N/A
2004/0050138	12/2003	Beck	N/A	N/A
2004/0194442	12/2003	Maertens	N/A	N/A
2007/0044445	12/2006	Spicer et al.	N/A	N/A
2007/0228707	12/2006	Curtis	N/A	N/A
2007/0289281	12/2006	Altepost	N/A	N/A
2008/0046154	12/2007	Bares	N/A	N/A
2009/0192734	12/2008	Mackin	N/A	N/A
2010/0089178	12/2009	Tragesser	73/862.627	A01D 43/085
2011/0173942	12/2010	Kowalchuk	N/A	N/A
2012/0029757	12/2011	Kowalchuk	N/A	N/A
2012/0042618	12/2011	Lohrentz	N/A	N/A
2012/0042619	12/2011	Lohrentz et al.	N/A	N/A
2012/0055131	12/2011	Zegota	N/A	N/A
2012/0055133	12/2011	Lohrentz	N/A	N/A
2012/0204528	12/2011	Regier	N/A	N/A
2012/0253611	12/2011	Zielke	N/A	N/A
2013/0125800	12/2012	Andphair et al.	N/A	N/A
2013/0152535	12/2012	Roberge	N/A	N/A
2014/0020354	12/2013	Tilly	N/A	N/A
2014/0116077	12/2013	Pierce	N/A	N/A
2014/0230391	12/2013	Hendrickson	N/A	N/A
2014/0230580	12/2013	Dybro	N/A	N/A
2014/0236381	12/2013	Anderson	N/A	N/A
2014/0311113	12/2013	Bonefas	N/A	N/A
2014/0331631	12/2013	Sauder	N/A	N/A

2015/0082760	12/2014	Zentner	56/62	A01D 45/021
2015/0082780	12/2014	Dueckinghaus	N/A	N/A
2015/0257337	12/2014	Schrattenecker	N/A	N/A
2015/0289438	12/2014	Sauder et al.	N/A	N/A
2015/0293029	12/2014	Acheson	N/A	N/A
2015/0293068	12/2014	Acheson et al.	N/A	N/A
2015/0302305	12/2014	Rupp	706/46	G06N 7/06 A01D 75/00
2015/0319929	12/2014	Hendrickson	33/504	
2015/0327440	12/2014	Dybro	N/A	N/A
2015/0334920	12/2014	Schleicher	N/A	N/A
2016/0037709	12/2015	Sauder et al.	N/A	N/A
2016/0041803	12/2015	Markov	N/A	N/A
2016/0084813	12/2015	Anderson	N/A	N/A
2016/0084987	12/2015	Dybro	N/A	N/A
2016/0120129	12/2015	Verhaeghe	N/A	N/A
2016/0174465	12/2015	Kohls	N/A	N/A
2016/0338267	12/2015	Anderson et al.	N/A	N/A
2016/0345485	12/2015	Acheson	N/A	N/A
2017/0061211	12/2016	Hundley et al.	N/A	N/A
2017/0089742	12/2016	Bruns	N/A	N/A
2017/0199528	12/2016	Detweiler	N/A	B64U 10/13
2017/0228118	12/2016	Sugumarn et al.	N/A	N/A
2017/0251600	12/2016	Anderson	N/A	N/A
2017/0332551	12/2016	Todd	N/A	N/A
2017/0339827	12/2016	Anderson	N/A	N/A
2018/0017965	12/2017	Kosa	N/A	N/A
2018/0084725	12/2017	Ostermeier	N/A	N/A
2018/0092303	12/2017	Goering	N/A	N/A
2018/0164471	12/2017	Dybro	N/A	N/A
2018/0199509	12/2017	Ricketts	N/A	N/A
2018/0228086	12/2017	Cook	N/A	N/A
2018/0338423	12/2017	Lucca	N/A	N/A
2018/0368321	12/2017	Noonan	N/A	N/A
2018/0373259	12/2017	Aberle	N/A	N/A
2019/0000007	12/2018	Schleicher	N/A	N/A
2019/0059223	12/2018	Seiders	N/A	N/A
2019/0110394	12/2018	VanNahmen	N/A	N/A
2019/0174667	12/2018	Gresch	N/A	N/A
2019/0195762	12/2018	Brune	N/A	N/A
2019/0261561	12/2018	Heitmann	N/A	N/A
2020/0000034	12/2019	Schlipf	N/A	N/A
2020/0008351	12/2019	Zielke	N/A	N/A
2020/0053961	12/2019	Dix	N/A	N/A
2020/0068803	12/2019	Sauder et al.	N/A	N/A
2020/0326674	12/2019	Palla	N/A	N/A
2020/0394580	12/2019	Bull	N/A	N/A
2021/0051849	12/2020	Asebedo	N/A	N/A

2021/0059114	12/2020	Eichhorn	N/A	N/A
2021/0132618	12/2020	Van Roekel	N/A	N/A
2021/0168991	12/2020	Dix	N/A	N/A
2021/0195824	12/2020	Van Roekel	N/A	N/A
2021/0315160	12/2020	Zielke	N/A	N/A
2021/0318118	12/2020	Eichhorn	N/A	N/A
2021/0321567	12/2020	Sidon	N/A	N/A
2021/0329837	12/2020	Schnaider	N/A	N/A
2021/0329838	12/2020	Zielke	N/A	N/A
2021/0352847	12/2020	Hunt	N/A	N/A
2022/0000024	12/2021	Zielke	N/A	N/A
2022/0071093	12/2021	Risius	N/A	N/A
2022/0078975	12/2021	Slichter	N/A	N/A
2022/0132737	12/2021	Anderson	N/A	N/A
2022/0225569	12/2021	Zielke	N/A	N/A
2022/0232759	12/2021	Sauder	N/A	N/A
2022/0317688	12/2021	Li	N/A	N/A
2022/0386527	12/2021	Schleicher	N/A	N/A
2023/0000015	12/2022	Herrmann	N/A	N/A
2023/0073551	12/2022	Holoubek	N/A	N/A
2023/0139169	12/2022	Cleodolphi	N/A	N/A
2023/0189690	12/2022	Friedlein	N/A	N/A
2023/0229163	12/2022	Rust	N/A	N/A
2023/0240185	12/2022	Zielke	N/A	N/A
2023/0243693	12/2022	McClelland	N/A	N/A
2023/0292664	12/2022	Zielke	N/A	N/A
2023/0389473	12/2022	Reed	N/A	N/A
2023/0401703	12/2022	Friedlein	N/A	N/A
2024/0023482	12/2023	Zielke	N/A	N/A
2024/0065156	12/2023	Woodcock	N/A	N/A
2024/0081171	12/2023	Schwartz	N/A	N/A
2024/0090379	12/2023	Eichhorn	N/A	N/A
2024/0373785	12/2023	Risius	N/A	N/A
2024/0389494	12/2023	Vorobiev	N/A	N/A
2025/0185540	12/2024	Eichhorn	N/A	N/A
2025/0194459	12/2024	Roe	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2020201559	12/2019	AU	N/A
1023764	12/2016	BE	N/A
102017018379	12/2017	BR	N/A
2313376	12/1999	CA	N/A
1928065	12/1969	DE	N/A
112014000906	12/2014	DE	N/A
2191439	12/2008	EP	N/A
2047738	12/2008	EP	N/A
3146831	12/2016	EP	N/A
2944179	12/2016	EP	N/A

3245859	12/2018	EP	N/A
3095313	12/2018	EP	N/A
3972381	12/2021	EP	N/A
6216795	12/2016	JP	N/A
2013078328	12/2012	WO	N/A

OTHER PUBLICATIONS

Farm Show, “Easton Goers, Inc. Space Wheel Checks Seed Placement In Field—Space Cadet”, 1996, Publisher: Farm Show. cited by applicant

D. Easton, “Com Population and Plant Spacing Variability: The Next Mapping Layer”, 1996, pp. 723-727, Publisher: ASA-CSSA-SSSA, Published in: Madison, WI. cited by applicant

Hummel et al., “Sensing Corn Population—Another Variable in the Yield Equation”, available as early as 2016. cited by applicant

Gore, L. M. 1996. Report: Stalk counter for VRT study fall of 1995. Deere & Co. Moline, IL, available as early as 2016. cited by applicant

Baweja, Harjatin & Parhar, Tanvir & Mirbod, Omeed & Nuske, Stephen. (2018). StalkNet: A Deep Learning Pipeline for high-Throughput Measurement of Plant Stalk Count and Stalk Width. cited by applicant

Birrell et al., “Com Population Sensor for Precision Farming”, “American Society of Agricultural Engineers. Annual Meeting”, 1995, vol. 95, No. 1334, Publisher: ASAE. cited by applicant

Jonathan P. Kelly, “By-Plant Prediction of Corn (*Zea mays* L.) Grain Yield Using Height and Stalk Diameter”, 2009, Publisher: Faculty of the Graduate College of the Oklahoma State University. cited by applicant

Haizhou Li, “Design and Evaluation of a Non-Intrusive Corn Population Sensor”, “Trace: Tennessee Research and Creative Exchange, Doctoral Dissertations, Graduate School”, Aug. 2007, Publisher: University of Tennessee, Knoxville. cited by applicant

Luck et al., “Sensor Ranging Technique for Determining Corn Plant Population”, “Faculty Papers and Publications In Animal Science—Animal Science Department”, 2008, vol. 920, Publisher: University of Nebraska—Lincoln. cited by applicant

Plattner, C. E., J. W. Hummel 1996. Corn Plant Population Sensor for Precision Agriculture. In: P.C. Robert, R.H. Rust, W.E. Larson, editors, Precision Agriculture, ASA, CSSA, SSSA, Madison, WI. p. 785-794. cited by applicant

Yeyin Shi, “Corn Plant Location, Spacing and Stalk Diameter Measurements Using Optical Sensing Technologies”, May 2014, Publisher: Faculty of the Graduate College of the Oklahoma State University. cited by applicant

Shrestha et al., “Automatic Corn Plant Population Measurement Using Machine Vision”, “Transactions of the ASAE”, 2003, pp. 559-565, vol. 46(2), Publisher: American Society of Agricultural Engineers. cited by applicant

Sudduth, K.A., Birrell, S.J., Krumpelman, M.J., Robert, P., Rust, R.H., & Larson, W.E. (2000). Field evaluation of a corn population sensor. cited by applicant

Sudduth et al., “Field Evaluation of a Corn Population Sensor”, available as early as 2016. cited by applicant

Drago, Row by Row 2020 Special Report Harvest Study Reveals Most Corn Heads Leaving Yield in the Field. cited by applicant

Headsight, Inc., “Row Guidance for Corn”, Jan. 2021, <https://headsight.com/row-guidance-corn>. cited by applicant

John Deere SSA, “AutoTrac RowSense | Precision AG | John Deere SSA”, Jan. 2021, <https://www.deere.com/sub-saharan/en/technology-products/precision-ag/autotracsense/>. cited by applicant

Reichhardt Electronic Innovations, “PSR Sensor Guidance”, Jan. 2021,

www.reichhardt.com/us_products_autosteer-system_autoguidance-tac.html. cited by applicant
Yu, H. Zheng, S. R. Kulkarni, H. V. Poor, "Outlier elimination for robust ellipse and ellipsoid fitting," in Proc. 3rd IEEE Int. Workshop Comput. Adv. Multi-Sensor Adapt. Process. (CAMSAP), Aruba, Dutch Antilles, Dec. 2009, pp. 33-36. cited by applicant
Contributions to Wikimedia projects (2020, Feb. 20), Numerical integration. Wikipedia.
https://web.archive.org/web/20200325094046/https://en.wikipedia.org/wiki/numerical_integration
(Year: 2020). cited by applicant
Area of a circle (Mar. 10, 2020). Wikipedia.
https://web.archive.org/web/20200325064526/https://en.wikipedia.org/wiki/Area_of_a_circle (year 2020). cited by applicant
Miller et al (Nov. 16, 2016). The Plant Journal.
<https://onlinelibrary.wiley.com/doi/pdf/10.1111/tpj.13320>. cited by applicant
Of GIS Ag Maps (here in as GIS), "Yield Monitor Data Post-Calibration (Linear and Non-Linear) Examples", Jan. 20, 2022, 8 pages + 1 SS page (9 pgs total). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. application Ser. No. 17/345,598, filed Jun. 11, 2021, now U.S. Pat. No. 11,758,845, which is a continuation of U.S. application Ser. No. 16/445,161, filed Jun. 18, 2019, now U.S. Pat. No. 11,064,653, that claims the benefit under 35 U.S.C. § 119(e) to U.S. Provisional Application 62/686,248, filed Jun. 18, 2018, and entitled "Corn Head Stalk Sensor" and U.S. Provisional Application 62/810,231, filed Feb. 25, 2019, and entitled "Corn Head Stalk Sensor Data Visualization," both of which are hereby incorporated herein by reference in their entirety for all purposes.

TECHNICAL FIELD

(1) This disclosure relates generally to agricultural implements, more particularly agricultural implements and sensors for detecting, measuring and displaying information about plant stalks during harvest.

BACKGROUND

(2) Generating accurate yield maps is an important tool in agriculture because it can assist stakeholders in making decisions and assessing prior actions. Prior yield maps analyze yields but do not detect missing plants or late emerged plants. Accurate knowledge of missing or late emerged plants may be useful to a practitioner in assessing the best course of action to improve future yields.
(3) There is a need in the art for devices, systems and methods for counting, measuring, and displaying data related to plant yield on a row-by-row basis during harvest.

BRIEF SUMMARY

(4) Disclosed herein are various harvesters, more specifically corn heads and associated sensors and data visualization systems on combine harvesters. Various sensors mounted on a corn head may count and measure corn stalks as they pass through the corn head during harvest. Processing components and display units are used to calculate and display information about the measured stalks to provide the user with information about yield including on a row-by-row and plant-by-plant level.

(5) A system of one or more computers can be configured to perform particular operations or

actions by virtue of having software, firmware, hardware, or a combination of them installed on the system that in operation causes or cause the system to perform the actions. One or more computer programs can be configured to perform particular operations or actions by virtue of including instructions that, when executed by data processing apparatus, cause the apparatus to perform the actions.

(6) One Example includes an agricultural system, including at least one stalk sensor assembly, where the at least one stalk sensor assembly is constructed and arranged to measure stalk size. Other embodiments of this Example include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

(7) Implementations according to this Example may include one or more of the following features. The agricultural system where the at least one stalk sensor assembly includes one or more wheels. The agricultural system where: the one or more wheels engage with stalks so as to rotate, and the agricultural system estimates the size of the stalks via the wheel rotation. The agricultural system where the one or more wheels is operationally coupled to one or more pulse sensors. The agricultural system where the one or more wheels is operationally coupled to one or more brakes. The agricultural system where the one or more wheels is operationally coupled to one or more proximity sensors. The agricultural system further including a visualization system. The agricultural system where the visualization system includes a user interface on an in-cab display. Implementations of the described techniques may include hardware, a method or process, or computer software on a computer-accessible medium.

(8) One general Example includes an agricultural system, including at least one stalk sensor assembly including: at least one wheel; at least one resilient member operatively engaged with the at least one wheel, at least one pulse sensor in communication with the wheel and constructed and arranged to detect degrees of rotation of the at least one wheel, where the degrees of rotation of the at least one wheel are correlated to stalk size. Other embodiments of this Example include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

(9) Implementations according to this Example may include one or more of the following features. The agricultural system further including a visualization system in communication with the at least one sensor assembly. The agricultural system where the visualization system includes a yield monitor. The agricultural system where the yield monitor is constructed and arranged to determine yield on a row-by-row basis. The agricultural system where the visualization system is constructed and arranged to calculate and display one or more of harvested plants, late emergence, missing plants, yield per plant, yield per area, yield per thousand, bushels per acre, bushels per thousand, economic loss, row-by-row area counting, and row plugs. The agricultural system further including a late emerged threshold defined by wheel rotation. Implementations of the described techniques may include hardware, a method or process, or computer software on a computer-accessible medium.

(10) One Example includes an agricultural system, including: a sensor assembly including: a first wheel and a second wheel, a first pulse sensor in communication with the first wheel, and a second pulse sensor in communication with the second wheel, where the first pulse sensor and the second pulse sensor measure degrees of rotation of the first wheel and the second wheel. Other embodiments of this Example include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

(11) Implementations according to this Example may include one or more of the following features. The agricultural system where the sensor assembly is constructed and arranged to estimate a stalk circumference from the degrees of rotation. The agricultural system further including a first resilient member operatively engaged with the first wheel and a second resilient member

operatively engaged with the second wheel. The agricultural system further including a visualization system. The agricultural system further including at least one proximity sensor. The agricultural system further including a row unit plug alarm. Implementations of the described techniques may include hardware, a method or process, or computer software on a computer-accessible medium.

(12) While multiple embodiments are disclosed, still other embodiments of the agricultural system will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative embodiments of the system and associated devices and methods of use. As will be realized, these systems, methods and devices are capable of modifications in various obvious aspects, all without departing from the spirit and scope of the disclosure. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side view of a row of corn plants.
- (2) FIG. 2 shows ears of corn from a row having missing plants.
- (3) FIG. 3 show ears of corn from a row without missing plants.
- (4) FIG. 4 is a side view of a row of corn plants with a late emerging plant.
- (5) FIG. 5 is a side view of a row of corn plants with late emerging plants.
- (6) FIG. 6 is a flow chart for troubleshooting low yield areas.
- (7) FIG. 7 is a top view of a harvester in a field, according to one implementation.
- (8) FIG. 8 is a top view of a harvester, according to one implementation.
- (9) FIG. 9A is a process diagram showing the operation of the sensory assembly, according to one implementation.
- (10) FIG. 9B is a model implementation of a sensor assembly, according to one implementation.
- (11) FIG. 9C is a model graph showing total rotation per unit time, according to the implementation of FIG. 9B.
- (12) FIG. 10A is a bottom view of a sensor assembly, according to one implementation.
- (13) FIG. 10B is a graph showing total rotation per unit of time, according to the implementation of FIG. 10A.
- (14) FIG. 11A is a bottom view of a sensor assembly, according to one implementation.
- (15) FIG. 11B is a graph showing total rotation per unit of time, according to the implementation of FIG. 11A.
- (16) FIG. 12A is a bottom view of a sensor assembly, according to one implementation.
- (17) FIG. 12B is a graph showing total rotation per unit time, according to the implementation of FIG. 12A.
- (18) FIG. 13A is a bottom view of a sensor assembly, according to one implementation.
- (19) FIG. 13B is a graph showing total rotation per unit time, according to the implementation of FIG. 13A.
- (20) FIG. 14A is a bottom view of a sensor assembly, according to one implementation.
- (21) FIG. 14B is a graph showing total rotation per unit time, according to the implementation of FIG. 14A.
- (22) FIG. 15A is a bottom view of a sensor assembly, according to one implementation.
- (23) FIG. 15B is a graph showing total rotation per unit time, according to the implementation of FIG. 15A.
- (24) FIG. 16 is a graph showing revolutions per minute per unit time, according to one implementation.
- (25) FIG. 17 depicts an exemplary user interface, according to one implementation

- (26) FIG. 18A depicts an exemplary user interface showing missing plant data, according to one implementation.
- (27) FIG. 18B depicts an exemplary user interface showing missing plant data, according to one implementation.
- (28) FIG. 19 shows exemplary harvest data.
- (29) FIG. 20 shows exemplary harvest data for various hybrids.
- (30) FIG. 21 shows exemplary harvest data for various hybrids.
- (31) FIG. 22 is a top view of a row unit, according to one implementation.
- (32) FIG. 23 is a top view of a corn head and row units, according to one implementation.
- (33) FIG. 24 is a side view of a row unit with plant stalk jam, according to one implementation.
- (34) FIG. 25 is a flow chart depicting an exemplary algorithm for row alarms, according to one implementation.
- (35) FIG. 26 is a side view of a row unit with a slow turning rate, according to one implementation.
- (36) FIG. 27 is a side view of a row unit with correct turning rate, according to one implementation.
- (37) FIG. 28 is a top view of a planter and a harvester comparing row numbers, according to one implementation.
- (38) FIG. 29 is an exemplary user interface, according to one implementation.

DETAILED DESCRIPTION

- (39) Various implementations of the disclosed systems, devices and methods relate to agricultural yield and monitoring systems having at least one of a data visualization system and any associated sensor assemblies. In some implementations, a harvester has a corn head comprising a plurality of row units. Each row unit or some of the row units have an operatively engaged stalk sensor assembly for counting and measuring plant stalks. The sensor assemblies are in communication with the data visualization system and associated computer(s) and/or other processing mechanisms to calculate and display various data including yield per acre, number of plants planted per acre, the number of plants harvested per acre, the percentage of missing plants, the percentage of emerged late plants, the yield per 1000 plants, the potential lost yield, economic loss per acre, and various other parameters as are herein disclosed and as would be recognized by those of skill in the art.
- (40) In various of these and other implementations, the sensor assemblies are rotational stalk sensor assemblies. These sensor assemblies mechanically engage plant stalks to detect and measure the plant stalks on an individual plant and row-by-row basis.
- (41) Turning to the drawings in greater detail, FIG. 1 depicts a row 1 of corn plants 2 having a missing plant (shown at 3 in FIG. 2). In an illustrative example, each plant 2 has a per plant yield equivalent to 200 bushels per acre (bu/ac). Plants 2 on each side of the missing plant 3 also have a projected per plant yield equivalent to about 200 bu/ac, illustrating no yield compensation for any missing plants 3. In some instances, plants 2 on either side of the missing plant 3 may marginally compensate for the missing plant 3 by growing a slightly larger ear. It is understood that any corresponding yield regain—by adjacent plants 2—is small or negligible. Most consider any yield compensation to be financially insignificant or negligible.
- (42) On a prior art yield map, such as that known in the art, missing corn plants 3 are interpreted or depicted as merely a lower yield for an entire field area. That is, areas with missing plants 3 can appear the same on a yield map as areas without missing plants 3 but as having smaller ears or lower yield.
- (43) FIGS. 2-3 show ears 4 taken from two different four-foot row strips in the same field. Three to four plants are missing in FIG. 2, while all plants are present in FIG. 3. One of skill would appreciate that the ears are smaller in FIG. 3 than the ears in FIG. 2. Consequently, both strips have the same yield quantity despite the greater number of ears in FIG. 3. These two strips would be indistinguishable on a yield map because their yield quantities are interpreted as the same, because there is not an accurate count of the missing plants 3.

(44) It will be further appreciated by those of skill in the art from FIG. 2-3 that knowledge of missing plants **3** is required to correctly troubleshoot low yielding areas on a yield map. That is because actions to correct missing corn plants **3**—no ears/no plant—are different from actions to correct low yields of a “full stand” row where all plants **2** are present but the ears are smaller.

(45) It is further understood that missing corn plants **3** are often caused by issues at planting that prevent or delay plant emergence. For example, mechanical planter problems like skips in planted seeds, improper seed depth, improper seed trench closure and crop residue that blocks plant emergence are various issues that may prevent plant emergence and lead to missing plants **3**. Further, agronomic factors at planting such as unviable seeds, cold soil temperatures, dry soil and soil diseases can also halt plant emergence and lead to missing plants **3**.

(46) By contrast, low yields from a full stand of corn are frequently caused by in season growing problems such as water shortage, fertilizer shortage, excessive heat during pollination, disease stress and other factors recognized by a person of ordinary skill in the art.

(47) Missing plants **3** commonly account for significant economic yield loss, sometimes as high as 10%. Additionally, the magnitude of missing plants **3** can vary substantially across fields. For example, the number of missing plants **3** can vary by soil type, slope as well as by different agronomic and tillage treatments.

(48) In addition to missing plants, late emerging plants can also significantly impact yield. Corn plants that emerge later than adjacent plants within a row typically do not match the size of the adjacent plants. Estimates used among those of skill in the art is about a 50% yield loss for plants behind by one leaf and about a 100% yield loss for plants behind two or more leaves. FIG. 4 depicts an exemplary corn plant **2A** behind more than two leaves. These late plants **2A** can be identified visually at harvest by their characteristic thin stalks and very small ears. FIG. 5 depicts late plants **2A** in the row of normal plants **2**. Empirically, emerged late plants **2A** are about half the size of thriving plants **2**. For example, late plants **2A** typically have about a 50% thinner stalk size than productive plants **2**.

(49) Late plants **2A** may be caused by many of the same factors that cause missing plants **3**, as discussed above (e.g., planting problems). For example, crop residue in the seed trench can stunt emergence by causing the corn shoot to have to grow around or through the residue. In another example, incorrect seed depth or seed trench closure may cause missing plants **3** or late plants **2A**. Planting practices may result in significant yield loss from emerged late plants **2A**, sometimes as high as 5% yield loss. As noted above, actions taken to correct or prevent late plants are often different than actions to correct a full stand low yield.

(50) Correctly assessing the cause of low yield is important for stakeholders to be able to determine the appropriate corrections that may be made to improve future yields. FIG. 6 is a flow chart depicting an exemplary implementation of a process tree for assessing and determining the cause of low yields in a planting environment. The ability to determine the number and location of missing plants **3** and late emerged plants **2A** within an area or entire field is addressed by the system **10** and associated devices and other components disclosed herein. This quantification and identification is valuable to assessing appropriate action(s) to improve future yields. Erroneous and/or economically wasteful corrective actions can be taken if missing plants **3** and emerged late plants **2A** are not considered when evaluating the cause of low yields. In one example, practitioners may assume a full stand and prescribe application of extra fertilizer in low yielding areas absent missing plant **3** or late emerging plant **2A** data. This extra fertilizer may be unnecessary if missing plants **3** from mechanical planter problems are the cause of the low yield, and resolving the mechanical issue is what is needed to improve yield. These and other types of errors may be prevented by considering missing plants and late emerged plants as a possible cause for low yields.

(51) Turning to FIG. 7, known yield monitors use a header **13** having a height sensor to trigger area counting on and off. That is, raising the head **13** above a certain height triggers area counting off and lowering the head **13** triggers area counting on. An issue can arise because the head **13** is not

always harvesting all rows **1** when it is down and therefore may over count an area. Various yield monitors may utilize an “auto swath” feature that automatically shuts off area counting for any corn head row over an already harvested area. Use of such an “auto swath” feature requires an expensive high accuracy GPS receiver to spatially detect which corn head row or rows should be shutoff.

(52) The disclosed agricultural system **10**, data visualization system **18**, sensor assemblies **20** and associated devices and methods address the issues described above by providing accurate stalk counts, stalk measurements, and corresponding data visualization. Those of skill in the art will readily appreciate that the various features described herein can be used independently or in combination.

I. Sensors

(53) FIG. **8** depicts an implementation of the system **10** having a harvester **11** with stalk sensor assemblies **20A-20H** disposed on each row unit **12** of a corn head **13**. The sensor assemblies **20** according to these implementations are constructed and arranged to detect and measure plant stalks **2**. For example, various implementations of the sensor assembly **20** are depicted in FIGS. **9B-15B**. In various of these and other implementations, the sensor assemblies **20** include rotational stalk sensors that are constructed and arranged to measure the perimeter of each stalk, or other stalk characteristics specific to the individual sensor type. In any event, these sensors **20** mechanically engage or otherwise interact with passing plant stalks to detect and measure plant stalks on an individual plant and row-by-row basis.

(54) FIG. **9A-9B** depict a model process diagram for an agricultural system **10** having a model sensor assembly **20**, according to certain exemplary implementations. FIGS. **9B, 10A, 11A, 12A, 13A, 14A** and **15A** depict various implementations and components of the sensor assembly **20**, while FIGS. **9C, 10B, 11B, 12B, 13B, 14B** and **15B** depict graphical representations of total rotation per unit time, according to their respective implementations.

(55) Looking at FIGS. **9A-9B**, FIG. **9A** depicts a process diagram and FIG. **9B** depicts certain components of an exemplary implementation of the system **10**.

(56) In these implementations, the sensor assembly **20** consists of two wheels **22** or other pulse generating devices. The wheels **22** engage with and rotate around a stalk **2**—as the stalk **2** enters and traverses through the corn head **13** in the direction of reference arrow **B**. In some implementations, the wheels **22** have teeth to grip and thereby engage with the stalk **2**. In various other implementations, the wheels **22** may have a smooth surface with a high friction or other gripping material such as rubber or the like disposed along the contact edges of the wheels **22**, in order to engage with the stalk **2**. In various implementations, the gripping material may be any type of rubber material, as would be appreciated by those of skill in the art.

(57) As the stalk **2** traverses through the corn head **13** the wheels **22** are engaged by the stalk **2** and the wheels **22** rotate about their axes. As the wheels **22** rotate a pulse sensor **24** or other sensor measures the amount of rotation of the wheel **22** via electrical or other pulses. In various implementations, a known amount of electrical pulses are symmetrically generated per revolution of the wheel **22**. For example, one electrical pulse may be generated for every 1/25 turn of the wheel **22**, such that twenty-five pulses will be emitted for each full revolution of the wheel **22**. In various other implementations—where the wheels **22** have teeth—the pulse sensor **24** may count the number of teeth that pass the sensor **24** per rotation. Various pulse sensors **24** are known and understood in the art. For example, the pulse sensor **24** may be an encoder, a gear teeth sensor, or other sensor as would be appreciated by those of skill in the art.

(58) It is further understood that in the various implementations of the sensor assembly **20**, the various components described herein are in operational communication with the visualization system **18** and/or any of the components described in relation to FIG. **8** that are capable of recording and storing digital or electronic information collected via the pulse sensors **26** and other components, as would be well-understood by those of skill in the art.

(59) In some implementations, the sensor assembly **20** is mounted under each stripper plate **30** in

front of the stalk rolls (shown in FIG. 22). In various alternative implementations, the sensor assembly **20** is mounted above the stripper plates **30**.

(60) Continuing with the implementations of FIGS. 9A-9B, the sensor assembly **20** operates such that a stalk **2** enters the corn head **13** (shown at reference arrow B) between the stripper plates **30** (box **100** in FIG. 9A), the stalk **2** engages the wheels **22** (box **102** in FIG. 9A) and the wheels **22** rotate and are urged angularly away from the stalk **2** (shown at reference arrow C) against a resilient member **26** (box **104** in FIG. 9A). The resilient member **26** allows the wheels **22** to move such that the stalk **2** can continue traverse through the stripper plates **30**.

(61) Continuing with FIGS. 9A-9B, the resilient members **26** also urge the wheels **22** toward the stalk **2**—such that the wheels **22** remain engaged with the stalk **2** and the wheels **22** are properly rotated about the entire circumference of the stalk **2**. Once the stalk **2** passes the wheels **22**, the wheels **22** are urged back to a return position by the resilient members **26** (box **106** in FIG. 9A).

(62) According to these implementations, as the stalk **2** is rotating the wheels **22**, one or more pulse sensors **24** are detecting and measuring the emitted electrical pulse from the wheel (box **108** in FIG. 9). The system **10** may then add the number of emitted pulses from the corresponding wheels **22** on both sides of the stalk **2**. The system **10** then can generate output corresponding to if the stalk **2**—that passed through the sensor assembly **20**—was a healthy stalk **2** or a late emerged stalk (box **112** in FIG. 9A).

(63) FIG. 9C depicts a graphical representation of the total rotation of the wheels **22** over time, which may be used to quantify or otherwise estimate the size of the stalk **2**, as discussed below. In use, it is thus understood that the wheels **22** start turning when a stalk **2** begins driving itself through the sensor assembly **20**. That is, the wheels **22** rotate around the circumference (or perimeter) of the stalk **2**. Each wheel **22** is constructed to emit pulses proportional to the degree of rotation, which are received by the pulse sensor **24**, as described above. These pulses and/or other signals of degrees of rotation are added from both wheels **22**—thereby detecting the entire circumference of the stalk **2**.

(64) Since stalks **2** are approximately circular, and healthy stalks **2** are larger in size than late emerged stalks **2A**, healthy stalks **2** produce significantly more degrees of rotation than late emerged stalks **2A**. Therefore, as discussed below in relation to Section VI, where the system **10** defines a late emerged threshold—such as a number of degrees of rotation (or pulses)—that threshold serves as a demarcation line for the system **10** to quantify and/or display via the visualization system **18** a quantification or other analysis of productive vs. late emerged stalks. Further implementations can utilize any of the collected data types or measurements in establishing and enforcing the late emerged threshold, such that, for example stalks above the threshold are recorded as productive and stalks below the late emerged threshold are recorded as late emerged by the system **10**. For example, stalks above 75° of wheel rotation are productive and below 75° are late emerged, as described further in relation to section VI below. Other implementations are of course possible, some of which utilize statistical techniques, machine learning and/or other artificial intelligence (AI) to establish and calibrate the late emerged threshold.

(65) Most stalks **2** are elliptical, meaning they have a major axis and a minor axis diameter, where major is a larger diameter than minor. Various prior stalk sensors only measure stalk diameter, but due to the elliptical nature of most stalks this type of measurement can introduce error depending on stalk orientation. The system **10** and sensor assembly **20** described herein eliminate this type of error by instead sensing the entire perimeter of a stalk.

(66) It is further understood that the wheels **22** may continue to rotate for a time after a stalk **2** passes, due to the angular momentum of the wheels **22**. This continued rotation can cause inaccurate measurements of the stalk **2** perimeter due to the continued rotation and detection of the rotation after the stalk **2** as entirely traversed the sensor assembly **20** and is no longer in contact with the wheels **22**. Various methods for addressing this are disclosed below.

(67) In some implementations, the sensor assembly **20** comprises brakes **28** that generate frictional

forces opposite the rotation of the wheels **22**. The brakes **28** when engaged rapidly reduce and stop the rotation of the wheels **22** when the stalk **2** is no longer in contact with the wheels **22**. The brakes **28** are configured to apply the correct amount of braking force when engaged to provide for a rapid deceleration of the rotation of the wheels **22** without causing slippage between the stalk **2** and the wheels **22**.

(68) FIGS. **10A-11B** depict further implementations of the system **10** comprising one or more brakes **28** to stop the rotation or counting of the rotation of the wheels **22** after the stalk has passed.

(69) In these implementations, the brake **28** or brakes **28** apply brake friction selectively based on position of the wheels **22**. In FIG. **10A**, the passing stalk **2** has urged the wheels **22** away from the brakes **28** such that no frictional force is being applied to the wheels **22**. The wheels **22** are rotating about the perimeter of the stalk **2** to record the stalk **2** size. FIG. **10B** depicts a graphical representation of the total number of rotations per unit of time that both wheels **22** have rotated about the stalk shown in FIG. **10A**. In FIG. **11A**, the stalk **2** has passed the wheels **22**, allowing the springs **26** to urge the wheels **22** into contact with the brakes **28**. The brakes **29** then apply frictional forces to halt the wheel **22** rotation, as shown in FIG. **11B**.

(70) Further implementations of the system **10** may adjust the frictional force applied based on system feedback. This feedback may include, but is not limited to, deceleration rate of the wheels **22** after a stalk **2** passes, vehicle ground speed, frequency of detected stalks **2**, and position of the wheels **22**.

(71) In various alternative implementations, as shown in FIGS. **12A-13B**, position sensors **32** are implemented in the sensor assembly **20**. These position sensors **32** are constructed and arranged to monitor the position and/or operation of the wheels **22**. For example, the position sensor **32** may monitor whether the wheels **22** have returned to a defined position indicating that the stalk **2** has passed. Additionally, the position sensors **32** may monitor when the wheels **22** have been urged away from the stalk **2** as the stalk **2** enters the sensor assembly **20**. This position information is then used to process the rotational signals emitted by the wheels **22** and sensed by the pulse sensors **24**. The position sensor **32** indicates when to start and stop counting rotation of the wheels **22**.

(72) FIGS. **12A-13B** demonstrate the use of such proximity sensors **32** to monitor position. In FIG. **12A**, the passing stalk **2** has forced the wheels **22** away from the proximity sensors **32**, such that rotations begin to be counted, as shown in FIG. **12B**. In FIG. **13A**, the stalk **2** has passed, allowing the springs **26** to urge the wheels **22** back into position adjacent to the proximity sensors **32**. FIG. **13B** shows when further rotation of the wheels **22** should be ignored due to the wheels **22** returning to a position engaged with the proximity sensors **32**. When the proximity sensors **32** are triggered, the system **10** ignores additional rotation signals until the proximity sensors **32** are again displaced.

(73) In another implementation, shown in FIGS. **14A-15B**, the resilient members **26** may be a resilient arm **26**. The resilient arm **26** is rigidly attached to the stripper plate **30** at a first end and the wheel **22** is mounted at the second end. In various implementations, the resilient arm **26** is made of an elastomer material, such as, but not limited to, polyisoprene, polybutadiene, polyisobutylene, and polyurethanes. The resilient member is constructed and arranged such that as the stalks **2** pass through the sensor assembly **20** the wheels **22** are urged out of the way causing the resilient member **26** to flex in the direction of reference arrow D. After the stalk **2**, completely traverses through the sensor assembly **20**, the resilient members **26** straighten or “snap” back into their original position.

(74) The implementation of FIGS. **14A-15B** may be used in conjunction with either or both of the brakes **28** and the proximity sensors **32**, discussed above.

(75) Various additional sensor implementations may employ the use of brakes **28** and/or proximity sensors **32** along with a maximum revolutions-per-minute (“RPM”) cutoff. After the stalk **2** passes through the sensor assembly **20**, the brakes **28** are engaged to slow the wheels **22**. As shown in FIG. **16**, the wheel(s) **22** reach a maximum RPM when the stalk **2** disengages from the wheels **22**. As such, a maximum RPM can be used to serve as an indicator for when to stop recording

additional rotation of the wheels **22**. It is understood that there is no single defined maximum RPM, instead the system stops reading/recording rotation when each wheel **22** reaches its peak or maximum RPM. The maximum RPM can vary from stalk to stalk according to ground speed and stalk size.

(76) In these and other implementations, the sensor assembly **20** begins recording degrees rotation when the proximity sensor **32** signals off-when a stalk **2** enters the sensor assembly **20**. The sensor assembly **20** continues to count and record rotations until the maximum RPM is reached. In various implementations, the RPM value is calculated by the pulse sensor **24**. The sensor assembly **20** then stops counting and recording rotation when the maximum RPM is reached. By only counting the revolutions prior to reaching the maximum RPM, the sensor assembly **20** may be more accurate in measuring stalk **2** perimeter because the sensor assembly **20** will not count the additional rotations while the wheels **22** are moving into contact with the brakes **28** or to engage with the proximity sensors **32**.

II. Data Visualization

(77) Turning back to FIG. **8**, in these and other implementations of the agricultural system **10**, the stalk sensor assemblies **20** are in communication with an electronic recording and visualization system **18**. In various implementations, the visualization system **18** has an in-cab display **15** and is interconnected with the sensor assemblies **20** through a wired or wireless connection **40**. In various implementations, the visualization system **18** communicates with a GPS receiver **42** and a yield monitor system **44**. It is appreciated that further hardware components are also in operational communication with these components and are constructed and arranged to effectuate the systems and processes described herein. That is, in various implementations, one or more processors **41** and physical storage media **43** are disposed in the harvester **11** or are otherwise accessible by the visualization system **18**, such as via a wired connection or a wireless connection such as an LTE or other cellular or Wi-Fi connection.

(78) It is further understood that the visualization system **18**, according to certain implementations, has or is otherwise connected to a server **45**, database **47** and other components necessary for calculation, processing, transmitting and otherwise storing data for use by the visualization system **18**, as described herein. Various of these components such as the server **45** and database **47** may be stored and accessed via a cloud **46** based platform. Alternate implementations comprise other hardware and software components necessary for effecting the processes described herein.

(79) Continuing with the exemplary implementation of FIG. **8**, in certain implementations, the various stalk sensor assemblies **20**, discussed above, may—in addition to counting and measuring plants **2**—detect late plants **2A**, plugged rows, and trigger area counting on/off. In some implementations, independent sensor assemblies **20** may be provided for each function: counting, plugging, area counting, and others functions as would be appreciated. It would also be appreciated that in some implementations, not every row unit **12** may require a stalk sensor assembly **20**. That is, in certain implementations, patterns unique to certain field scale conditions, like hybrid type, planting date, tillage, and the like can be detected by instrumenting only some of the corn head rows **12**. The system **10** may function with or without a yield monitor **44** or a GPS receiver **42**.

(80) Each stalk sensor assembly **20**, according to certain implementations, is assigned a row number to denote a row unit **12** of a corn head **13** (shown as **20A**, **20B**, **20C**, etc.). It is appreciated that by convention corn head **13** row units **12** are commonly numbered from left to right with respect to forward travel direction (shown as reference arrow **A**). Distance offsets may be used to locate the various sensors assemblies **20** relative to the mounted GPS receiver **42** location on the harvester **11**.

(81) In further implementations, each stalk sensor assembly **20** independently counts and measures stalks **2** entering the respective corn head **13** row unit **12** and may, in certain implementations, determine a quantity of harvested plants **2**, missing plants **3**, emerged late plants **2A** as well as row plugged and row area counting on/off status, among other characteristics of the plant/row. Further

implementations are of course possible.

(82) FIG. 17 for example depicts an exemplary user interface **19** of a row-by-row bar graph of on-the-go or instantaneous data from each stalk sensor assembly **20** (shown in FIGS. 8, 9B, 10A, 11A, 12A, 13A, 14A, and 15A at **20**) which may appear via the visualization system **18** on the in-cab display **15**. It should be appreciated that all stalk sensor assembly **20** data, such as the row-by-row bar graph data, can be visualized in many different formats. For example, the data may appear in the user interface **19**, as a numerical display or a row-by-row color map with a legend indicating magnitude of each parameter.

III. Harvested Plants

(83) Various implementations of the agricultural system **10** assess the quantity of harvested plants. Harvested plants are a count of corn stalks **2** harvested by each row unit **12** of the corn head **13**. The stalk sensor assembly **20** and associated system **10** may be constructed and arranged to allow a user to optionally exclude emerged late plants **2A** from harvested plants **2**. This ability is useful because, as discussed above, emerged late plants **2A** do not contribute any significant yield and therefore the technique allows for quantifying only the productive corn plants **2**.

(84) In some implementations, harvested plants **2** can be visualized as plants per area or as a percent of planted seeds per area. The harvested plant data may be displayed in real-time or near real-time on a row-by-row basis (shown for example in FIG. 17). The data can also be in numerical form or a color map with a legend indicating harvested plant magnitude. In various implementations, harvested plants **2** can be visualized as an average across all rows. The average may be expressed in different visualization forms, such as a bar graph, numerically, a color map with a legend indicating magnitude of harvested plants, and/or other forms as would be appreciated by those of skill in the art. In some implementations, the average harvested plants can be visualized and compared by treatment, such as by hybrid, tillage or planter treatments.

(85) In various implementations, the number of harvested plants **2** is interchangeable with the ear count. This is because nearly all corn stalks **2** only have one ear **4**, therefore the number of harvested plants **2** is typically about equal to the number of productive ears **4**. This relationship is more accurate in implementations wherein emerged late plants **2A** are excluded from the number of harvested plants **2**. In this disclosure, the use of ear count is interchangeable with the use of harvested plants **2**.

IV. Missing Plants

(86) The system **10**, according to certain implementations, is constructed and arranged to calculate, display, log and map missing plants **3** via the visualization system **18**. Counting and mapping missing plants **3** is valuable to various stakeholders because, as discussed above, corn plants **2** adjacent to missing plants **3** either do not compensate for the yield lost from a missing plant **3** or the compensation is negligible. For this reason, missing plants **3** often represent the largest economic impact of any parameter that the corn stalk sensor assembly **20**, discussed herein, measures.

(87) In some implementations, missing plants **3** can be visualized in real-time or near real-time on a row-by-row basis via the visualization system **18**, as shown in FIG. 17. The missing plant **3** data can also be in numerical form or a color map with a legend indicating missing plant magnitude. In these and other implementations, missing plants **3** can also be visualized as an average across all rows. In addition, the missing plants can also be visualized as a percent of missing plants based on different treatments, like planting date or hybrids, as shown in FIGS. 18A and 18B. The average can be expressed in different visualization forms, such bar graph, numerically or a color map with a legend indicating magnitude of missing plants **3**.

V. Quantification

(88) Various approaches to the visualization system **18** are configured for quantifying missing plants **3**, such as those discovered via the sensor assemblies **20**. According to one implementation, the system **10** may determine missing plants by subtracting stalks counted by the stalk sensor

assemblies **20** from the quantity of seeds planted where the harvester **11** is harvesting. For example, if the stalk sensor assemblies **20** count 30,000 harvested plants/ac in an area in which the planter planted 32,000 seeds/ac, the system **10** subtracts 32,000-30,000, and then displays and records 2,000 missing plants per acre. The planted seed quantity may be expressed as a seeding rate (number of seeds planted per area) and may come from “as-planted” information electronically and spatially recorded by the planter system.

(89) Farmers and other practitioners may have access to “as-planted” information during harvest by use of the same visualization system **18** and display **15** in the harvester **11** as the planter. In some implementations, the harvester **11** display **15** is different than the planter display and the “as-planted” information can be downloaded from a cloud storage system or any other file transfer means known to those of skill in the art. In various implementations, the “as-planted” seeding rate may be derived from planter seed sensor readings, a manually-entered target seeding rate into the planter system, or a prescription seeding rate map. In implementations where field areas are planted at a single seeding rate, the user can enter a target seeding rate into the system **10**.

(90) In alternate implementations, missing plants **3** can be calculated without user entered seeding rates or “as-planted” seeding rate information by using stalk sensor assemblies **20** to measure the average corn plant spacing at harvest. Corn planted at a fixed seeding rate has a theoretical target seed spacing. For example, corn planted at 32,000 seeds per acre at 30-inch row spacings are typically spaced about 6.5 inches apart. Planters are typically not capable of spacing every seed exactly 6.5 inches apart; however, average seed spacing within a field as measured by planter monitors are typically close to the target spacing (in this example 6.5 inches). Because seeds do not move in the soil, corn stalk sensor assemblies **20** can measure the space between every plant at harvest and calculate the same average seed spacing as the planter monitor.

(91) Stalk sensor assemblies **20** determine seed spacing by measuring the distance the harvester **11** travels between each counted stalk **2**. The system **10** according to certain implementations calculates distance traveled from a harvester **11** ground speed source, such as GPS, radar, transmission speed sensor, or other source known to those of skill in the art. In various implementations, the system **10** records a field average plant spacing and updates the value as new stalks are harvested and counted. The system **10** may then back calculate the original target seeding rate using the corn head **13** row unit **12** space setting and the stalk sensor assembly **20** average plant spacing.

(92) In one example, the target seeding rate (seeds/ac) is:

$$(93) \frac{6,272,640(\text{seeds})}{\text{averageplantspacing}(\text{inches}) \times \text{rowspacing}(\text{inches})}$$

(94) In another example, the target seeding rate (seeds/hectare) is:

$$(95) \frac{100,000000(\text{seeds})}{\text{averageplantspacing}(\text{cm}) \times \text{rowspacing}(\text{cm})}$$

(96) As described above, the calculated target seeding rate minus counted stalks calculates the missing plant quantity.

(97) In certain implementations, the system **10** visualization system **18** quantifies missing plants as plants per area or as a percent of planted seeds. For example, if the stalk sensor assemblies **20** count 30,000 harvested plants/ac in an area where the planter planted 32,000 seeds/ac, the system **10** calculates, displays and records 2,000 missing plants or stated another way 6.25% missing plants. Missing plants expressed on a percent basis can be a more useful metric in certain circumstances than plants/area. For example, different treatments like corn hybrids or planting date may characteristically produce a certain percentage of missing plants, regardless of the planted seeding rate. The ability to compare treatments on a percentage basis exposes the proportional tendency of the treatment.

(98) FIG. **18A** shows missing plant trends for various exemplary hybrid types. FIG. **18B** shows a proportional tendency for more missing plants at earlier planting dates compared to later planting dates, independent of the planted seeding rate. Other treatments that may be compared including,

but not limited to, planting depth, row unit down force, soil moisture, soil temperature, seed to soil contact, tillage depth, fertilizer rate, and others as would be appreciated by those of skill in the art.

VI. Late Emergence

(99) In some implementations, the agricultural system **10** measures the quantity of late emerged plants, as was also discussed above in relation to FIG. 9B-C. In certain of these implementations, late emergence is quantified by detecting the stalk size proximate the plant bottom, approximately the first 0-3 feet of stalk above ground. As discussed above, late emerged plants **2A** stalk size can be approximately half the stalk size of a productive plant **2**. The stalk size may be in terms of cross-sectional stalk area, circumference, or perimeter, as discussed above. Exemplary stalk sensor assemblies **20** measuring and distinguishing between late and productive plants are described throughout this disclosure, and the processing and display of collected data about late emergence can be achieved via the visualization system **18** implementations discussed variously herein such as in relation to FIG. **8**.

(100) Late emerged plants can be expressed, recorded, logged, mapped, displayed and otherwise visualized with the same units and user interface **19** techniques as described above with respect to missing plants, as would be readily appreciated by the skilled artisan.

VII. Yield Per Plant or Area

(101) In various implementations of the system **10**, corn yield can be measured in yield units per area (“YPA”), for example, bushels/acre or metric tons/hectare. Corn hybrids, fertilizer rates, seeding rates and many other agronomic and mechanical treatments are compared on a YPA basis. The corn stalk sensor assemblies **20**, in combination with a yield monitor **44** integrated with the operations system of the visualization system **18** are configured to measure yield characteristics, such as yield per plant. The yield per plant is typically a very small number, and as such it may be useful to express yield per 1000 plants or other value.

(102) As described herein, yield per 1000 plants may be expressed as YPK. The YPA may be expressed as the BPA or bushels per acre. The yield per 1000 plants in bushels may be expressed as BPK. In some implementations, yield per plant can be calculated from YPA, derived from the yield monitor **44**, divided by the quantity of harvested plants derived from the stalk sensor assemblies **20**.

(103) In a specific example, shown in FIG. **19**, a yield monitor **44** measures 250 bu/ac and stalk sensor assemblies **20** count 32,000 harvested plants/ac, not including any emerged late plants. The system **10** via the visualization system **18** and associated processing components, described above, may then divide 250 by 32,000 to achieve a yield per plant of 0.0078 bushels. In certain implementations the system **10**, may multiply that calculated yield per plant by 1000 to achieve of a yield per 1000 plants. In some implementations, the system **10** can divide the yield (250 bu/ac) by number of harvested plants divided by 1000, to equal about 7.8 bushels per 1000 plants.

(104) In certain implementations, yield per plant is visualized as ear weight. To calculate ear weight:

$$(105) \frac{YPA \times 56}{\text{HarvestedPlantQuantity}}$$

Where YPA is derived from the yield monitor **44** and the harvested plant quantity is derived from the stalk sensor assemblies **20**. A value of 56 pounds per bushel is a standard value for corn. The YPA is a volumetric measurement that must be converted to weight when yield per plant is calculated and visualized using ear weight.

(106) YPK and ear weight are expressed in different units (volume and weight respectively), the values can be used interchangeably when evaluating the proportional difference between various treatments.

(107) YPK may be a useful metric where it is important to visualize the yield with respect to only the number of harvested plants. YPK is a function of kernel weight and kernel count per plant of the plants present at harvest. YPA is a function of kernel weight and kernel count per plant but is lowered by yield lost due to missing and emerged late plants. In other words, YPA reflects a yield penalty for missing plants and emerged late plants.

(108) Because plants standing at harvest do not make up the yield lost from missing plants **3** or emerged late plants **2A** or any compensation is negligible, YPK can be useful for comparing the yield response of agronomic and/or mechanical treatments. YPK can represent the true agronomic yield response better than YPA in cases where a mechanical issue proportionally caused more missing plants **3** and/or emerged late plants **2A** in a particular treatment.

(109) FIG. **20** shows treatment comparison, specifically a corn hybrid comparison. Hybrid P 0987AMX yielded 13 bushels less than two DK hybrids on a YPA basis. However, Hybrid P 0987AMX yielded about the same on a YPK basis. The YPA difference can be attributed to the increased number of missing plants for the P 0987AMX hybrid, 9.2% of the P 0987 plants were missing at harvest versus about 2% for the DK hybrids. YPK of the harvested P 0987 plants is 7.7, similar to the YPK of the DK hybrids of 7.6. It is agronomically reasonable to assume that missing plants could yield about 7.7 bu/1000 plants had they been there at harvest. If the missing plant rate of the P 0987AMX hybrid was similar to that of the DK hybrids, the YPA of P 0987 would be equal to or greater than the YPA of the two DK hybrids.

(110) In reference to the above example, this data would let a farmer or other practitioner know that missing plants **3** and/or late emerging plants **2A** are likely an important factor in the lower yield of the P 0987AMX hybrid. As discussed in more detail above, mechanical problems with the planter may be the reason for a high number of missing plants **3** or emerged late plants **2A**. These missing plants **3** or emerged late plants **2A** cause a lower YPA of an otherwise high yielding hybrid. This information allows a farmer or other practitioner to make fully informed decisions. For example, knowing all three hybrids had the same YPK and noting a mechanical problem caused the P 0987 missing plants, means the farmer may now consider P 0987 to be equal in yield to the two DK hybrids if planted properly.

(111) FIG. **21** shows various yield data for certain exemplary corn hybrids. In this example, emerged late plants **2A** are excluded from harvested plants **2**. The YPA of both hybrids was the same at 173 bu/ac. Yet the YPK of the 42-98 hybrid was 20% higher than the 61-49 hybrid (7.1 compared to 5.9 bu/1000 plants). This information allows for assessing if a certain hybrid is higher yielding, has a chronic missing plant problem, or has independent mechanical causes for lower yields. This data can assist a farmer or other practitioner in assessing crop yield and determining the best hybrids for subsequent plantings, as would be understood.

(112) In various implementations of the system **10**, the stalk sensor assembly **20** and the visualization system **18** can calculate and visualize YPK and/or ear weight comparisons from different agronomic and/or mechanical treatments. The YPK or ear weight can be visualized as an average of across all row or in other ways as would be recognized. In various implementations, the data can appear in various forms including a bar graph, numerically, and/or a color map with a legend indicating YPK and/or ear weight magnitude via GUI in the in-cab display **15**. Alternate implementations utilize further data metrics and display techniques, as would be readily appreciated by those of skill in the art.

(113) It is understood that YPK and/or ear weight visualization via the visualization system **18** can be used for row-by-row yield monitoring. It is further understood that in various implementations, the stalk sensor assemblies **20** do not directly measure the amount of grain per stalk, but that the amount of grain per stalk can be estimated due to the strong correlation between the number of harvested stalks to the number of productive ears, as discussed above. That is, using the visualization system **18**, processing components, and previous data sets, it is possible to estimate or otherwise predict the grain per stalk given certain known or gathered parameters, such as an instantaneous or near real-time yield, number of stalks harvested per unit area and/or per row, YPK, and others as would be appreciated, each of which can be entered by the user, gathered via the system **10**, determined by the sensor assemblies **20** or pulled from stored data on a database **47**, server **45**, in the cloud **46** or elsewhere, as would be readily appreciated by those of skill in the art.

(114) In various implementations, the system **10**, and associated processing components, may

determine the row-by-row YPA according to the ratio of harvested productive plants, excluding missing plants and late emerged plants, in each row to the average counted harvested productive plants across all corn head rows. The system **10** can include an instantaneous, or near real-time yield monitor **44**, as described above. Additionally, the system **10**, in conjunction with the sensor assemblies **20**, can determine the number of stalks **2** harvested per row. In one specific example—wherein it is assumed that all productive plants harvested have the same YPK—the real time yield is 250 bu/ac and the row-by-row count of stalks **2** harvested is 30,000 plts/ac for row **1**, 25,000 plts/ac for row **2**, 30,000 plts/ac for row **3**, and 30,000 plts/ac for row **4**. The system **10** may divide the number of harvested plants per row by the average number of harvested plants across all rows to determine a per row plant ratio.

(115) Continuing with the above example, the per row plant ratio is 1.043 for row **1**, 0.870 for row **2**, etc. The plant ratio may then be multiplied by real-time yield to determine the YPA per row. In the example, the YPA for row **1** is 260, 218 for row **2**, etc.

(116) An exemplary equation for determining a YPA per row is:

$$(117) \frac{\text{Count}_n}{\text{Avg harvested plants across all corn head rows}} = \frac{\text{Yield}_n}{\text{Avg yield across all corn head rows}}$$

where n is the individual row number.

(118) In another implementation, the row-by-row yield can be calculated by multiplying the YPK by the number of harvested plants per acre by row (expressed in thousands). The YPK may be determined by dividing the yield (250 bu/ac) by the average number of harvested plants per acre (**28.75**). It would also be appreciated by those of skill in the art that other equations and methods, such as a direct proportional distribution method, may be used to determine row-by-row YPA values.

(119) In these and other implementations, the row-by-row YPA can be visualized via the visualization system **18**. The various data, including the row-by row YPA, can be visualized in different forms, for example, bar graph, numerically, or a color map with a legend indicating YPA magnitude. In some implementations, the data can be visualized comparatively by agronomic or mechanical treatment.

VIII. Economic Loss

(120) Various implementations of the system **10** having a visualization system **18** are constructed and arranged to calculate and display economic loss.

(121) It will be appreciated that YPA quantifies the amount of sellable grain in all field areas, but it does not quantify the yield “that could have been.” Said another way YPA does not quantify the potential yield lost due to missing plants **3** and emerged late plants **2A**.

(122) The missing plants **3** and/or emerged late plants **2A** can be expressed as an estimated economic or monetary loss to help practitioners comprehend the extent of financial loss better than plants per area and percent quantifications.

(123) In some implementations, the system **10** can perform various calculations, including determining missing plants **3** and emerged late plants **2A** in plants per area units averaged across all rows of the corn head **13**. FIGS. **20** and **21** show missing plants **3** and emerged late plants **2A** as a percent quantification. The system can calculate the number of missing plants **3** and emerged late plants **2A** by multiplying the number of planted plants/ac by the percentage of missing plants **3** and the percentage of emerged late plants **2A**. To calculate only one parameter (missing plants **3** or late emerged plants **2A**) the system **10** omits the other parameter percentage from the calculation.

(124) Using hybrid 42-98 in FIG. **21** an example the calculation may be:

$$(125) \frac{34,300 \text{ plant / ac} \times (25\% + 4\%)}{100} = 9,947 \text{ plants / ac}$$

As desired, if only the number of missing plants **3** was desired the 4% can be removed from the calculation, such that the number of missing plants **3** is 8,575 plants/ac.

(126) In various implementations the system **10** can determine potential lost yield. Potential lost yield can be calculated by multiplying the number of missing and emerged late plants divided by

1000, as calculated above, by the YPK. This value represents the estimated yield loss from missing plants **3** and emerged late plants **2A** assuming a 100% yield loss for the missing plants **3** and late emerged plants **2A**.

(127) Continuing with the above specific example, the potential yield loss is:

9.947 (plants/1000)/ac×7.1=71 bu/ac

(128) In some implementations, the system **10** can determine economic loss. For this calculation a selling price must be assumed or determined. The economic loss may be calculated by multiplying the potential lost yield, as calculated above by the selling price/bushel. The selling price can be user entered in the system **10** or otherwise gathered from an external or internal source.

(129) Continuing with the above specific example, the economic loss is:

71 bu/ac×\$3.50/bu=\$249/ac loss

(130) In this example the price per bushel is assumed to be \$3.50 per bushel.

(131) The various data can be visualized in different forms, for example, bar graph, numerically, or a color map with a legend indicating economic loss magnitude. In some implementations the data can be visualized comparatively by agronomic or mechanical treatment. For example, a bar graph, as shown in FIG. **18A** or FIG. **18B**, can be modified by substituting on the Y axis various calculated or measured parameters.

IX. Area Counting

(132) In some implementations, the system **10** is constructed and arranged for harvester area counting that is row independent, also referred to as row-by-row. In these and other implementations, the stalk sensor assemblies **20** allow for row-by-row area counting. In these and other implementations, each stalk sensor assembly **20** turns area counting on when it detects a stalk **2** feeding into the row unit **12**. Each stalk sensor assembly **20** turns area counting off when stalks **2** stop feeding into the row unit **12**. The area count contribution of each row is a function of harvester ground speed multiplied by row width, and can be calculated and displayed via the visualization system **18** as would be appreciated.

(133) As an exemplary implementation, FIG. **7** depicts an eight row, 30-inch row width corn harvester **11** finishing a pass. In this example, stalks **2** are no longer feeding into corn head **13** rows **1-3**. As such the stalk sensor assemblies **20** of rows **1-3** are not counting area, while the stalk sensor assemblies **20** of rows **4-8** are counting area as signaled by the presence of stalks **2** feeding into each row **12**.

(134) At various times one or more rows **12** may have an extended length of consecutive missing plants **3**—for example 3+ feet with no plant. In this situation, to maintain YPA accuracy, the system **10** keeps area counting on because the missing plants **3** represent an unintended economic loss.

(135) In some implementations, to prevent false area count off triggers, the system **10** can shut off area counting in a cascading fashion from the outside of the corn head **13** in. Said another way, an inside row will not shutoff until an adjacent more exterior adjacent row shuts off. For example, referring to FIG. **7**, row **3** will not shutoff until row **2** shuts off, and row **2** will not shutoff until row **1** shuts off. This outside to inside daisy chain or cascading type method prevents false area count off triggers on inside rows.

(136) In certain implementations, the two furthest outside row units **12** will not shut off until they experience more than a threshold amount of consecutive distance, such as feet or meters, of missing plants **3**. The threshold may be a user entered setting or a hard-coded setting, such as about 3 feet. Other distances ranging from about one inch to about 10 feet or more are of course possible in alternate implementations.

X. Row Alarms

(137) Certain implementations of the agricultural system **10** utilize row alarms. Referring now to FIG. **22**, corn heads **13** have two stalk rolls **58** per row **12** that pull and crumple the stalk down through the corn head **13** to the ground. A set of stripper plates **30A**, **30B** strip the ear off the stalk **2** as the stalk **2** is pulled down through a gap between the plates. Gathering chains **52A**, **52B** (also

referred to as gathering fingers) carry the stripped ears to a cross auger that conveys to the harvester feeder hose.

(138) Stalks, rocks, grass, weeds, soil, and the like may from time to time jam in the stalk rolls **58** or gathering chains **52A**, **52B** imposing a high torque on the stalk rolls **58** and gathering chains **52A**, **52B**. To prevent mechanical breakage, corn head row units **12** in some implementations include a mechanical clutch that will slip at high torques. In these implementations, when the clutch starts to slip, the row unit **12** RPM slows and the row unit **12** may move in a jerky motion. The row unit **12** may stop turning altogether if it cannot clear the jam through the stalk rolls **58** and gathering chains **52A**, **52B**.

(139) Turning to FIGS. **23** and **24**, if the row unit **12** stops turning, stalks **2** can jam up next to each other in front of the row unit **12**. These plugged rows may result in a 100% yield loss until they are unplugged because the row unit **12** stops gathering ears altogether. As a further complication, in certain situations it is understood that after the plugged row jams full of stalks **2**, the row unit **12** pushes over and breaks off stalks leaving all the ears on the ground.

(140) In use, according to certain implementations, the agricultural system **10** may detect a plugged row unit **12** by sensing one or more stalks **2** jammed together via the sensor assembly **20** with proximity sensor **32**. During corn head **13** operation, the sensor assembly **20** according to these implementations is configured to detect gaps or spaces between individual stalks **2** via the proximity sensor **32** as the stalks **2** enter the row unit **12** and pass through the sensor assembly **20**. For example, the proximity sensor **32** can measure the stalk gap distance by measuring the time or distance between when the proximity sensor **32** is on—when the wheels **22** return to their original position—to when the proximity sensor **32** is off indicating the next stalk has entered the sensor assembly **20**. The stalk gap distance can be set to a defined threshold, such that when the stalk gap distance violates the defined threshold a jam has occurred. Accordingly, the system **10** according to these implementations is configured to issue a row plugged alarm, such as on the in-cab display **15**, when no stalk gaps are detected for a defined travel distance or time, or the stalk gap distance otherwise violates the defined threshold. Other parameters can be used in alternate implementations, as would be understood.

(141) There are further possible complications; often corn heads **13** have too many rows for an operator to effectively visually watch for jams at typical harvest speeds. Additionally, outside corn head **13** rows may be difficult to see due to their distance and orientation away from the cab of the harvester **11**. The visual difficulty may increase due to nighttime harvests or dust.

(142) FIG. **25** is flowchart depicting an exemplary algorithm for a row plug alarm **120** within the system **10** based on distance traveled since the last stalk gap was detected. In various implementations, the threshold distance may be entered by a user or may be a predefined setting within the system **10**. In various implementations, the distance threshold may be about 3 feet and the gap threshold distance may be about 0.5 feet. It is readily appreciated that other threshold distances and gap threshold distances may be employed in alternate implementations.

(143) In various implementations, the row plug alarm **120** will be issued after a series of steps are executed via the system **10** and associated hardware components, discussed above. During harvest the sensor assemblies **20** detect the presence of stalks **2** (box **122**), as discussed above. The row plug alarm **120** then asks if there is a stalk gap within the gap threshold distance (box **124**). If there is a stalk gap within the threshold distance, no plug is detected, no alarm is issued and the sensor assembly **20** begins sensing the next stalk **2** and stalk gap.

(144) If the sensor assembly **20** does not detect a stalk gap or the stalk gap is less than the threshold distance, the system **10** may then add the distance traveled since the last detected stalk to the distance counter (box **128**). Next, the system will ask if the distance counter is greater than the distance threshold (box **130**). If the distance counter is less than the distance threshold, no plug is detected, no alarm is issued and the sensor assembly **20** begins sensing the next stalk **2** and stalk gap. If the distance counter is greater than the distance threshold then a plug is detected and the row

plug alarm is issued (box 132).

(145) As shown in FIG. 26, row units 12 may turn slow and cause stalks 2 to bunch together as they feed into the row unit 12. This slow turning and bunching may cause header grain loss through uneven feeding or plugging. Row units 12 turning too fast can cause a high ear impact on stripper plates (shown in FIG. 22 at 30A, 30B) that shell grain off the ear bottom. Corn header row units 12 need to be turning fast enough that stalks 2 maintain their plant spacing as they feed into the head 13 and but not so fast as to cause high impact forces on the stripper plates 30A, 30B, as shown in FIG. 27.

(146) Farmers or other operators may set the corn head 13 speed to correctly feed stalks 2 for a certain ground speed. As conditions change and operators change harvester ground speed, the corn header 13 RPM needs to be adjusted as well, but often is not. In various implementations, the stalk sensor assemblies 20 can detect if the corn head 13 speed is either too fast or too slow for the current ground speed by looking for gaps between stalks 2 that are different than gaps that correspond to the target seeding rate. The system 10 can issue an alarm to alert an operator that the row unit 12 is not operating at the correct or optimal speed. The alarm may continue until the sensor assembly 20 detects gaps that correspond to the target seeding rate. In some implementations, the alarm may notify the harvester operator to decrease travel speed or increase the header RPM or a combination of both. The operator knows the travel speed and/or header RPM is corrected when the alarm ceases.

(147) In alternate implementations, the system 10 may automatically control header speed by the rate of stalks 2 passing by the sensor assembly 20. In one specific example, the optimum corn head speed for 10 stalks per second entering the row unit 12 may be 450 RPM and 15 stalks per second might be 525 RPM. The system 10 may automatically adjust the corn head speed to 525 RPM when the rate jumps from 10 to 15 stalks per second. In various implementations, the automatic speed control can be independent, row-by-row, or for all rows together.

XI. Planter Rows

(148) The system 10 may have access to “as-planted” spatial information recorded from the planting operation. Using the harvester 11 GPS location, the system 10 determines the as-planted pass the harvester 11 is harvesting. Because planter rows 12 commonly exceed corn head 13 rows by a multiple of two or three, the system 10 must find the right set of contiguous planter rows within the planter pass width that aligns with the corn head 13 row units 12. Various planter recording systems spatially record each row and its planter row number. In these and other implementations, the system 10 uses the harvester 11 GPS location to associate the correct set of planter row numbers to the corn head 13.

(149) The corn head 13 to planter row association can be automatically reset for every harvested pass as triggered by when the harvester 11 starts harvesting a new pass as shown in FIG. 28.

(150) In implementations where the as planted information does not contain planter row numbers, the system 10 may determine the planter row number to corn head 13 row number association by comparing planting direction to the harvester 11 direction and taking into account planter width and where the harvester 11 lines up within that planter width.

(151) The system 10, in some implementations, may record and display the field average harvested plants 2, missing plants 3 and emerged late plants 2A per area by each planter row number. The system 10 may update field averages as new stalks 2 are harvested and counted.

(152) FIG. 29 depicts an exemplary in-cab display 15 and user interface 19. The real-time or near real-time feedback of data from the sensor assemblies 20 can indicate mechanical issues with the planter. For example, in FIG. 29, the fourth planter row unit has a field average of 15% missing plants, which is substantially higher than other row units, which indicates something may be wrong with the fourth row unit and/or fourth row.

(153) In some implementations, the system 10 has a reset or restart function for the planter row average. The reset or restart function can be a manual or automatic function. The system 10 may

reset or restart the data collected when harvesting a new condition or field section. In one example, an automatic reset function may be triggered when the harvester **11** switches to a new field, field sub-region or hybrid.

(154) In another example, the system **10** may trigger an automatic reset or restart when the harvester **11** enters a field area that has a different pre-recorded planter, sprayer, fertilizer or tillage parameter, such as seed depth, closing wheel adjustment, row cleaner adjustment, row unit gauge wheel down force, or other parameters as would be recognized by those of skill in the art.

(155) In various implementations, the system **10** names and records each instance as a numerical summary of the average performance of each planter row.

(156) Although the disclosure has been described with reference to various embodiments, persons skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the disclosed apparatus, systems and methods.

Claims

1. An agricultural data visualization system for use on a corn head comprising one or more row units, the data visualization system comprising: (a) one or more sensors disposed on the one or more row units, each of the one or more sensors configured to sense stalks passing through the corn head; (b) a GPS receiver in association with the corn head configured to detect a location of the corn head during harvest operations; (c) a display in communication with the one or more sensors and the GPS receiver; and (d) a storage system comprising as-planted spatial information in communication with the display, wherein the display is configured to correlate the as-planted spatial information with the location of the corn head during harvest operations, wherein each of the one or more row units is correlated to a planter row unit on the basis of the as-planted spatial information, and wherein correlation of the one or more row units and the planter row unit is reset for every harvester pass.
2. The system of claim 1, wherein the GPS receiver is at a known offset from each of the one or more sensors, and wherein the known offset is used to map locations of one or more sensors during harvest operations.
3. The system of claim 1, wherein the one or more sensors are configured to measure at least one of a number of harvested plants, a number of missing plants, seed spacing, and stalk size.
4. The system of claim 1, wherein the display comprises a user interface for display of data from the one or more sensors on-the-go.
5. The system of claim 4, wherein data from the one or more sensors is displayed on a row-by-row basis.
6. A stalk detection system for an agricultural harvester comprising one or more row units comprising: (a) at least one stalk sensor assembly disposed on each of the one or more row units, where the each of the at least one stalk assemblies are configured to detect plant stalks; (b) a processor; and (c) a GPS receiver in electronic communication with the agricultural harvester configured to determine locations of the agricultural harvester during harvest operations, wherein the processor is configured to compare detected plant stalks to a number of expected stalks based on planting data and wherein the processor is configured to use the locations of the agricultural harvester during harvest operations to match harvested rows with planter row numbers.
7. The system of claim 6, wherein the processor is configured record and display one or more of a number of plants planted row-by-row, a number of plants harvested row-by-row, a percentage of missing plants per row, a number of missing plants per row, and a percentage of emerged late plants per row.
8. The system of claim 6, wherein the at least one stalk sensor assembly is configured to detect stalk size.

9. The system of claim 8, wherein the system is configured to detect late emerged plants based on detected stalk size.
